



3rd year

إشارات ونظم معالجة
د. جواد شالطة

Signals & Systems Lab

Lab (3)

Types of Signals (Continuous)

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Categorization by the Variable Type Continuous-Time Signals



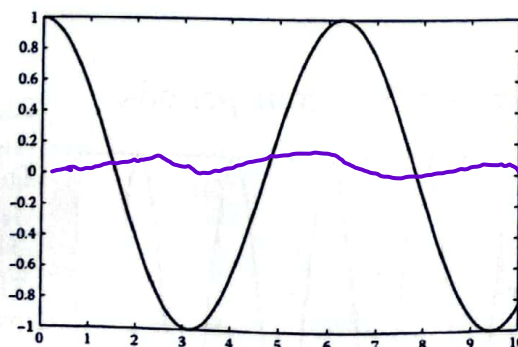
continuous-time (or analog)

independent variable (time) defined in a continuous interval.

variable t is continuous.

The dependent value that usually denotes the amplitude of the signal is also a continuous variable.

step
`t = 0:0.01:10`
`y = cos(t);`
`plot(t,y)`



Time (the independent variable t) is defined by using a very small step (time step = 0.01) in the continuous domain $0 \leq t \leq 10$.

The dependent variable $y(t)$ is defined in the continuous set of values $-1 \leq y(t) \leq 1$.

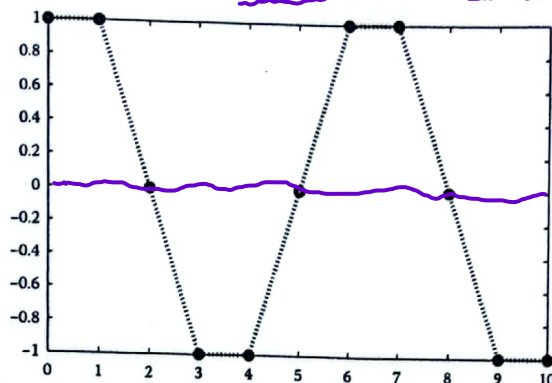
The analog signal is drawn by using the command `plot`

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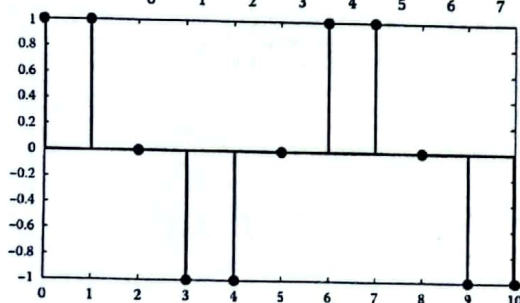
Digital Signals

Digital signals are the signals that both independent and dependent variables take values from a discrete set.

```
n=0:10;
y=cos(n);
y=round(y);
plot(n,y,'o')
```



```
stem(n,y)
```



Using the command `stem` is more appropriate also for digital signals.

Discrete time n (independent variable) is defined with step 1.

Definition of the dependent variable $y[n]$. The variable $y[n]$ is rounded toward the nearest integer; that is, it takes values from a discrete set.

Graph of the discrete-time signal by using the command `plot` with the proper syntax. The points determined by the circles are the values of the variable $y[n]$.

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Basic Continuous-Time Signals

2.2.1 Sinusoidal Signals

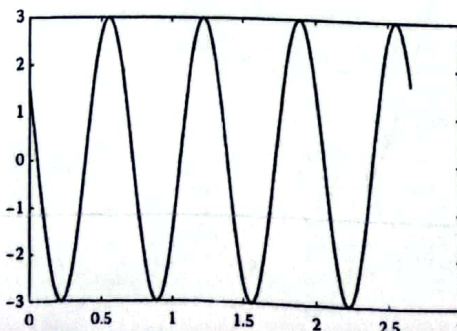
$$x(t) = A \cos(\Omega t + \theta)$$

The first basic category presented is that of sinusoidal signals. This type of signal is of the form $x(t) = A \cos(\Omega t + \theta)$, where Ω is the angular frequency, given in rad/s, A is the amplitude of the sinusoidal signal, and θ is the phase (in radians). Sinusoidal signals are periodic signals with fundamental period T given by $T = 2\pi/\Omega$ s. Finally, a useful quantity is the frequency f given in Hertz. Frequency f is defined by $f = 1/T$ or $f = \Omega/2\pi$.

Example

Plot the signal $x(t) = 3 \cos(3\pi t + \pi/3)$ in four periods. $\rightarrow 2\pi T$

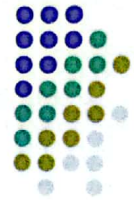
```
A=3;
Omega=3*pi;
thita=pi/3;
T=2*pi/Omega;
t=0:0.01:4*T;
x=A*cos(Omega*t+thita);
plot(t,x)
```



The amplitude of the signal is 3.
The angular frequency is 3π .
The phase is $\pi/3$.
The period is $2/3$.
The time is defined from 0 to $4T$.

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Basic Continuous-Time Signals

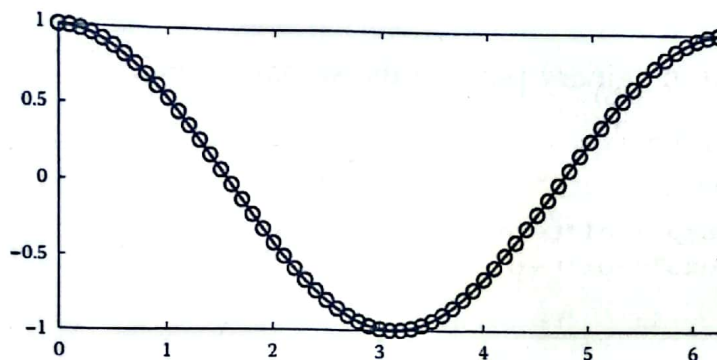


2.2.1 Sinusoidal Signals

Commands

```
t=0:0.1:2*pi;  
x1=cos(t);  
x2=sin(t+pi/2);  
plot(t,x1,t,x2,'o')  
xlim([0 2*pi])
```

Results



Comments

It is clear that the signals $\cos(t)$ and $\sin(t + \pi/2)$ are identical. The signal $\cos(t)$ is plotted with the solid line, while $\sin(t + \pi/2)$ is plotted with the circles. One can say that cosine is a sine with phase $\theta = \pi/2$.

Extend plot to be to the limit values !!

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Exponential Signals



$$x(t) = Ae^{bt}$$

$b > 0$, increasing function

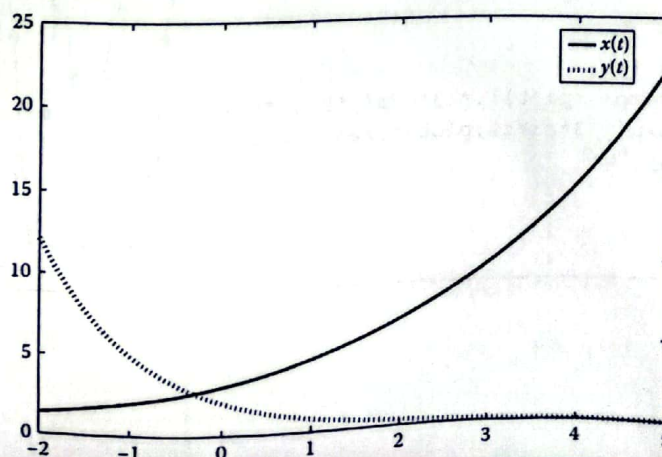
$b < 0$ decreasing function

Example

Plot the signals $x(t) = 3e^{0.4t}$ and $y(t) = 2e^{-0.9t}$ in the time interval $-2 \leq t \leq 5$.

```
t=-2:.1:5;  
x=3*exp(0.4*t);  
y=2*exp(-0.9*t);  
plot(t,x,t,y,':');  
legend('x(t)', 'y(t)')
```

Important



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Complex Exponential Signals

$$Ae^{j\Omega t + \theta} = A(\cos(\Omega t + \theta) + j\sin(\Omega t + \theta)). \quad \text{Euler's formula}$$

$T = 2\pi/\Omega$ fundamental period

$$\operatorname{Re}\{Ae^{j\Omega t + \theta}\} = A\cos(\Omega t + \theta) \quad \operatorname{Im}\{Ae^{j\Omega t + \theta}\} = A\sin(\Omega t + \theta)$$

exponential

adjective · UK / ek.speɪ.nen.ʃl/ US / ek.spoʊ.nen.ʃl/

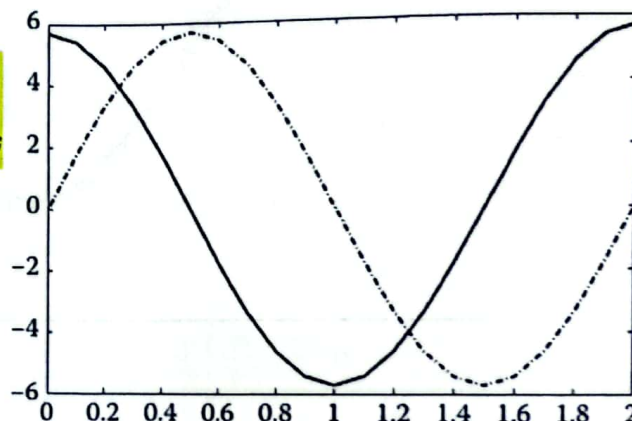
Example

Plot the real and imaginary parts of the signal $y(t) = 2e^{j\pi t + \pi/3}$ in time of one period.

$$T = 2\pi/\Omega = 2\pi/\pi = 2.$$

```
t=0:0.1:2;
y_re=real(2*exp(j*pi*t+pi/3));
y_im=imag(2*exp(j*pi*t+pi/3));
plot(t,y_re,t,y_im,'-.');
```

The real part of $y(t)$ is $2\cos(\pi t + \pi/3)$ while the imaginary part of $y(t)$ is $2\sin(\pi t + \pi/3)$.



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Complex Exponential Signals

$$Ae^{j\Omega t + \theta} = A(\cos(\Omega t + \theta) + j\sin(\Omega t + \theta)).$$

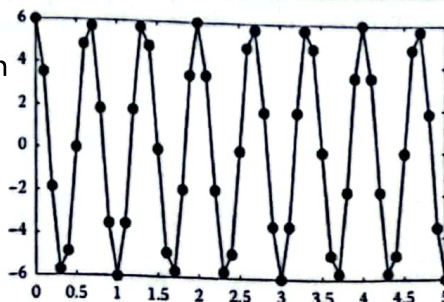
Example

Plot the real parts of the signals $x(t) = 2e^{j\pi t}$ and $y(t) = 6e^{j3\pi t}$.

Commands	Results	Comments
----------	---------	----------

```
t=0:0.1:5;
x=(2*exp(j*pi*t)).*(3*exp(j*2*pi*t));
y=6*exp(j*3*pi*t); plot(t,real(x),t,
real(y),'ko')
```

new operation



The real parts of $x(t)$ and $y(t)$ are the same.

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Unit Step Function

$$u(t) = \begin{cases} 1, & t > 0 \\ \frac{1}{2}, & t = 0 \\ 0, & t < 0 \end{cases}$$

For convenience $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$

heaviside(t)

heaviside

Heaviside step function

Syntax

heaviside(x)

Description

heaviside(x) returns the value 0 for $x < 0$, 1 for $x > 0$, and $1/2$ for $x = 0$.

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Unit Step Function

```
heaviside(sym(-3))
```

```
ans =  
0
```

```
heaviside(0)
```

```
ans =  
0.5000
```

```
syms x  
ezplot(heaviside(x), [-2, 2])
```



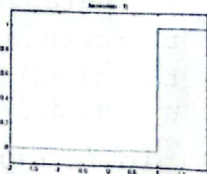
```
heaviside(sym(3))
```

```
ans =  
1
```

```
syms x  
assume(x < 0)  
heaviside(x)
```

```
ans =  
0
```

```
ezplot(heaviside(x - 1), [-2, 2])
```



```
heaviside(sym(0))
```

```
ans =  
1/2
```

```
syms x clear
```

```
syms x  
heaviside(sym([-1 0; 1/2 x]))  
  
ans =  
[ 0,      1/2]  
[ 1, heaviside(x)]
```

```
syms x  
diff(heaviside(x), x)  
  
ans =  
dirac(x)
```

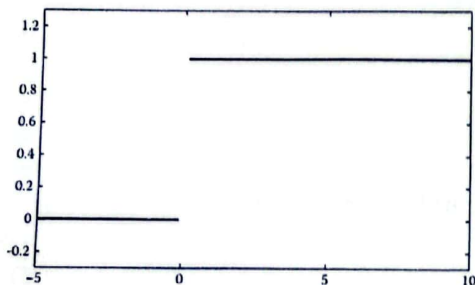
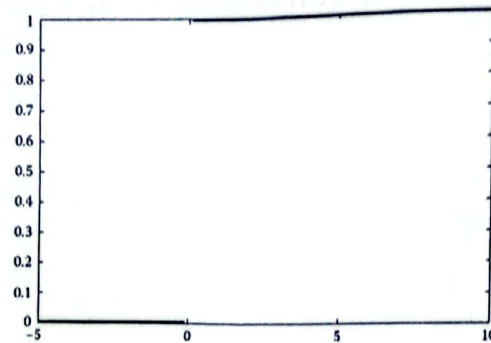
```
syms x  
int(exp(-x)*heaviside(x), x, -Inf, Inf)  
  
ans =
```

Unit Step Function

First Method

With use of the command heaviside.

```
t=-5:0.1:10;
u=heaviside(t)
plot(t,u)
ylim([-0.3 1.3])
```



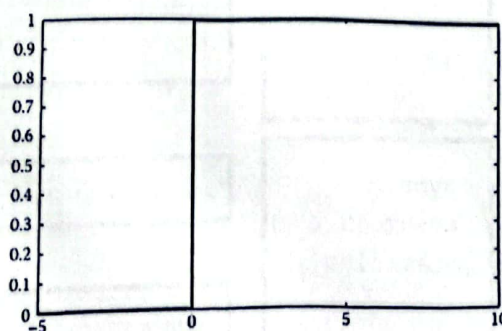
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Unit Step Function

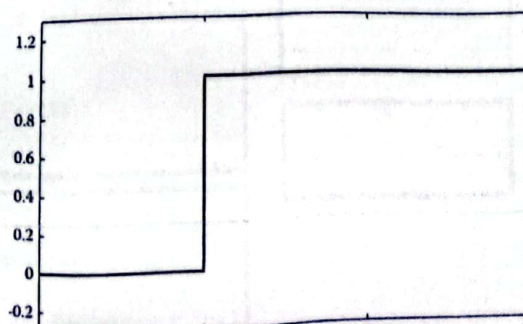
Second Method

Use of the technique of defining and plotting piecewise functions.

```
t1=-5:.1:0
t2=0:.1:10
u1=zeros(size(t1))
u2=ones(size(t2))
t=[t1 t2];
u=[u1 u2];
plot(t,u);
```



```
ylim([-0.3 1.3])
```



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Unit Step Function

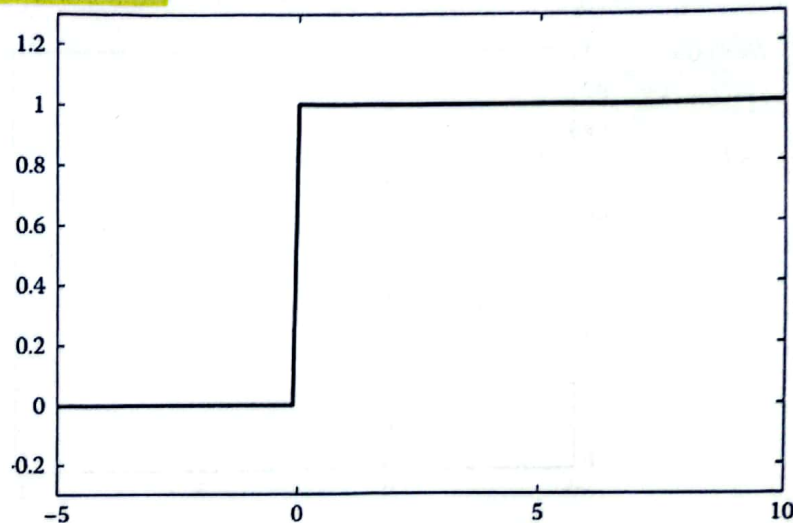
Third Method

Implementation with specific number of zeros and ones.

```
t = -5:.1:10;
```

```
u = [zeros(1,50) ones(1,101)];
```

```
plot(t,u);  
ylim([-0.3 1.3])
```



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Unit Impulse or Dirac Delta Function

The Dirac function $\delta(t)$

$f(\cdot)$ is an arbitrary function

$$\int_{-\infty}^{\infty} f(t)\delta(t) = f(0).$$

$$f(t) = 1, t \in (-\infty, \infty).$$

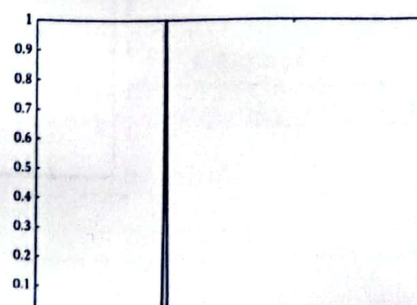
$$\int_{-\infty}^{\infty} \delta(t) = 1$$

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

dealing with discrete-time signals

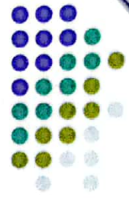
$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad \begin{array}{l} \text{Delta or the} \\ \text{Kronecher function} \end{array}$$

```
t1 = -5:.1:-0.1;  
t2 = 0;  
t3 = 0.1:.1:10;  
d1 = zeros(size(t1));  
d2 = 1;  
d3 = zeros(size(t3));  
t = [t1 t2 t3];  
D = [d1 d2 d3];  
plot(t,D)
```



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Unit Impulse or Dirac Delta Function

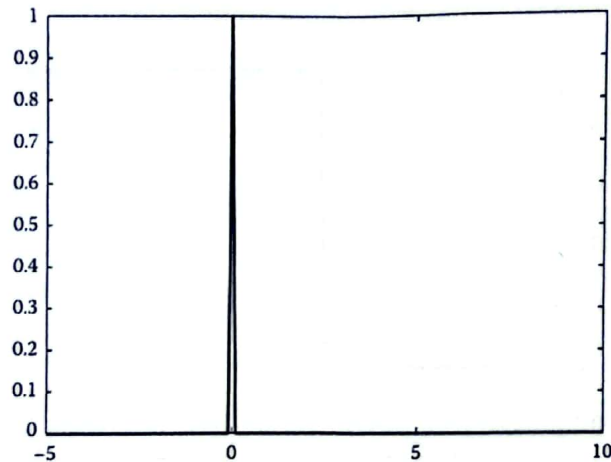


dealing with discrete-time signals

$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad \begin{array}{l} \text{Delta or the} \\ \text{Kronecher function} \end{array}$$

The command `gauspuls` is not created in MATLAB for that purpose but it is convenient for us to use it in the special way that is presented in this book.

```
t=-5:.1:10;  
s=gauspuls(t)  
plot(t,s)
```



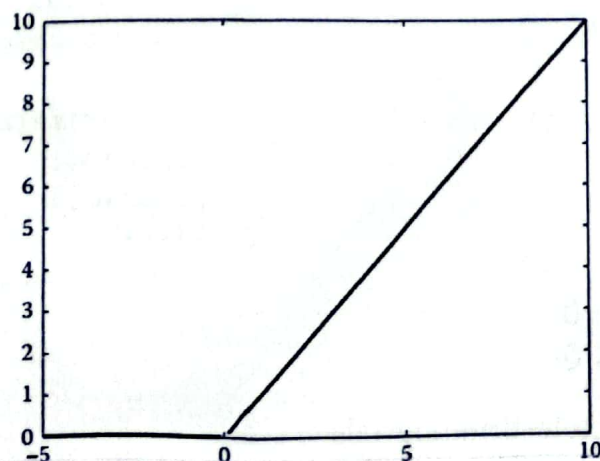
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Ramp Function

$$r(t) = t \cdot u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

The unit-ramp function $r(t)$

```
t=-5:0.1:10;  
r=t.*heaviside(t);  
plot(t,r)
```



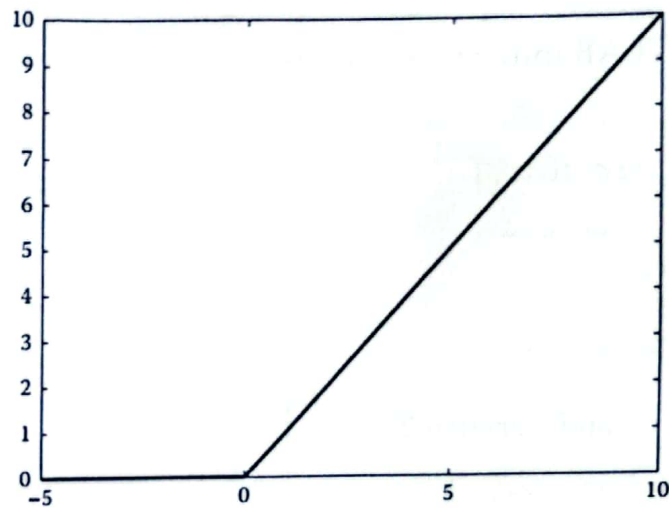
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Ramp Function

$$r(t) = t \cdot u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

The unit-ramp function $r(t)$
technique of piecewise
functions

```
t1=-5:.1:-0.1;
t2=0:.1:10;
r1=zeros(size(t1));
r2=t2;
t=[t1 t2];
r=[r1 r2];
plot(t,r)
```



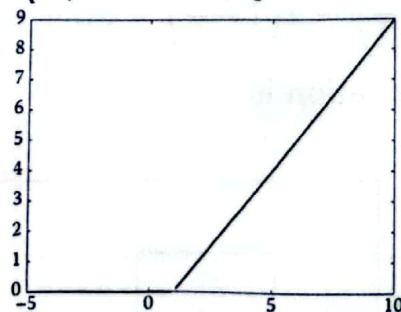
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Ramp Function

The general form of the (unit) ramp function is

$$r(t - t_0) = (t - t_0)u(t - t_0) = \begin{cases} t - t_0, & t \geq t_0 \\ 0, & t < t_0 \end{cases}$$

```
t=-5:.1:10;
r=(t-1).*heaviside(t-1)
plot(t,r)
```



Commands

Results

Comments

`syms t`

Definition of the symbolic variable t .

`u=heaviside(t)`

`u=heaviside(t)`

Definition of $u(t)$.

`int(u,t)`

`ans = heaviside(t)*t`

$\int u(t)dt = tu(t) = r(t)$.

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Rectangular Pulse Function

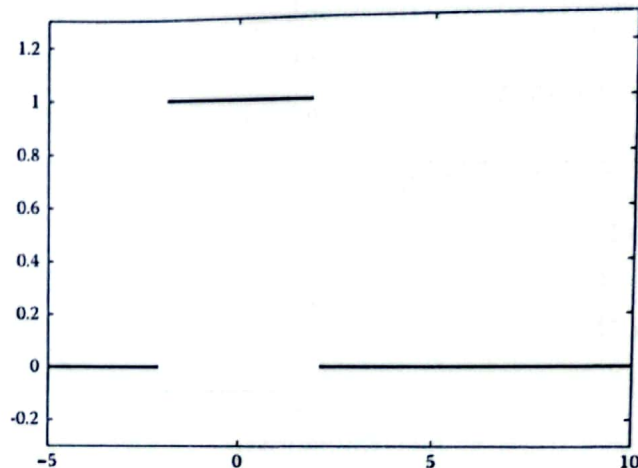
$$p_T(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right) = \begin{cases} 1, & -T/2 \leq t \leq T/2 \\ 0, & \text{elsewhere} \end{cases}$$

$$T=4 \quad p_4(t) = u\left(t + \frac{4}{2}\right) - u\left(t - \frac{4}{2}\right) = u(t+2) - u(t-2).$$

The MATLAB implementation is

```
t=-5:.1:10;
u1=heaviside(t+2);
u2=heaviside(t-2);
p=u1-u2;
plot(t,p)
ylim([-0.3 1.3])
```

unit magnitude and duration $T=4$
centered at $t=0$.



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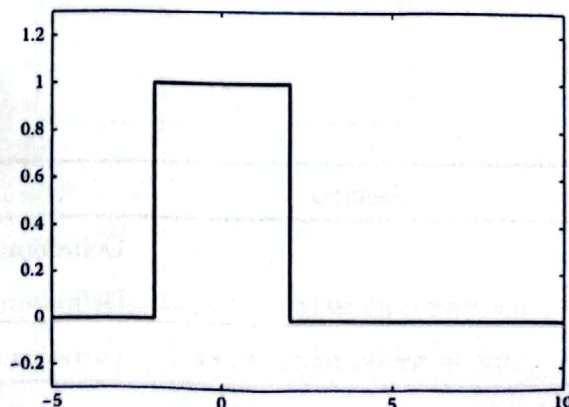
Rectangular Pulse Function

$$p_T(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right) = \begin{cases} 1, & -T/2 \leq t \leq T/2 \\ 0, & \text{elsewhere} \end{cases}$$

$$T=4 \quad p_4(t) = u\left(t + \frac{4}{2}\right) - u\left(t - \frac{4}{2}\right) = u(t+2) - u(t-2).$$

The MATLAB implementation is

```
t1=-5:.1:-2;
t2=-2:.1:2;
t3=2:.1:10;
p1=zeros(size(t1));
p2=ones(size(t2));
p3=zeros(size(t3));
t=[t1 t2 t3];
p=[p1 p2 p3];
plot(t,p);
ylim([-0.3 1.3]);
```



Graph of the signal and
modification of the y-axis
for better appearance.

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Rectangular Pulse Function

$$p_T(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right) = \begin{cases} 1, & -T/2 \leq t \leq T/2 \\ 0, & \text{elsewhere} \end{cases}$$

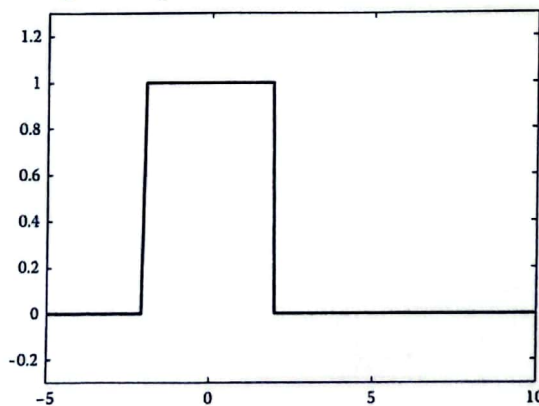
$$T=4 \quad p_4(t) = u\left(t + \frac{4}{2}\right) - u\left(t - \frac{4}{2}\right) = u(t+2) - u(t-2).$$

The MATLAB implementation is

`rectpuls(t)` get a rectangular pulse of duration $T=1$

`rectpuls(t,a)` rectangular pulse of duration $T=a$.

```
t=-5:.1:10;  
s=rectpuls(t,4);  
plot(t,s)  
ylim([-0.3 1.3])
```



Graph of the signal
and modification
of the y-axis for
better appearance.

Thank you